EXPERIMENT - 24

24. Implement the Black hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

Aim

To implement the Black Hat technique as a morphological operation using OpenCV in Python to highlight small dark regions in an image.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Resize the image to fit the screen.
4. Convert the image into grayscale.
5. Create a kernel using np.ones().
6. Apply the Black Hat operation using cv2.morphologyEx().
7. Display the original image and the Black Hat output image.

Program

```
import cv2
import numpy as np

img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
else:
    img = cv2.resize(img, (500, 350))
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kernel = np.ones((5, 5), np.uint8)

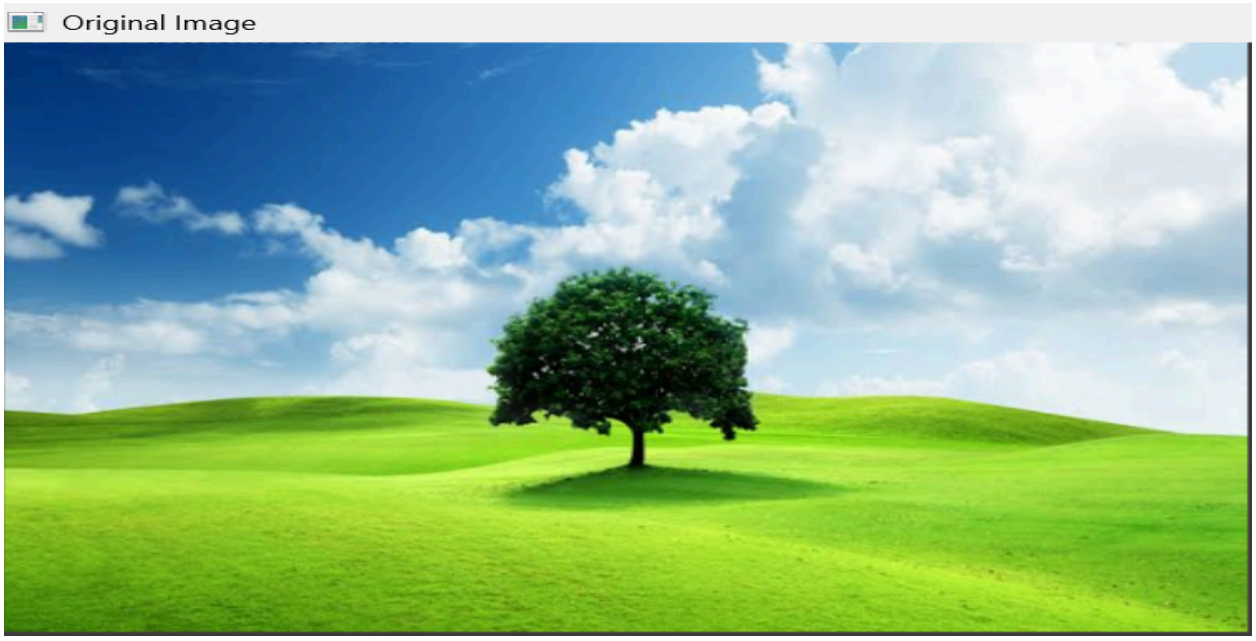
    blackhat = cv2.morphologyEx(gray, cv2.MORPH_BLACKHAT, kernel)

    cv2.imshow("Original Image", img)
    cv2.imshow("Black Hat Output", blackhat)

    cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Sample Input and Output



Output Image



Result

The Black Hat Morphological Operation was successfully implemented using OpenCV in Python.